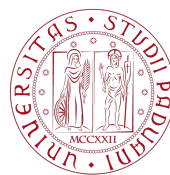


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**UNIVERSITÀ
DEGLI STUDI
DI PADOVA**

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RESEARCH PROJECT

Dvoretzky Theorem

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Chapter 1

Convex Bodies and Norms

In this chapter, we lay the groundwork for Dvoretzky's theorem: we formalize a concept of distance between Banach spaces, and we discuss the geometric intuition behind this. We then discuss the Banach-Mazur compactum, the concept of dual of balls and work on some functional analysis' lemmas that will help us "move" the geometry from one space to another. We conclude by recalling some properties of standard Gaussian measures.

We will mostly work with finite-dimensional spaces.

1.1 Balls and Gauges

In this section we work on finding a correspondence between norms and convex bodies.

Definition 1.1. Let $B \subset \mathbb{R}^n$. We call B a *ball* if B is a compact, convex, symmetric (with respect to the origin) subset of $\mathbb{R}^n$ with a non empty interior.

Notice that to each norm on $\mathbb{R}^n$ we can associate a ball just by taking its unit ball. We now show that this correspondence is bijective.

Definition 1.2. We define the *gauge* of B as the map

$$\begin{aligned} \|\cdot\|_B : \mathbb{R}^n &\rightarrow [0, +\infty) \\ x &\mapsto \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x \in \lambda B\}. \end{aligned}$$

Proposition 1.3. Let B be a ball. Then the gauge of B $\|\cdot\|_B$ is well defined and a norm with unit ball B .

Proof. We first notice that $\|\cdot\|_B$ is well defined. There exists $\epsilon > 0$ such that $B(0, \epsilon) \subset B$. This follows from the non empty interior and symmetry hypotheses: $\exists x_0 \in B$, $\epsilon > 0$ such that $B(x_0, \epsilon) \subset B$. By symmetry $B(-x_0, \epsilon) \subset B$ and by convexity we get $B(0, \epsilon) \subset B$. We conclude by noticing that

$$x \in B\lambda \quad \forall \lambda \in \left(\frac{2\|x\|}{\epsilon}, \infty\right).$$

We now prove that $\|\cdot\|_B$ is a norm. We observe that $\|x\|_B = 0$ iff $x = 0$. Indeed, if $x = 0$, $x \in 0B$ so $\inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x \in \lambda B\} = 0$. Conversely, if $\|x\|_B = 0$ we get that $\exists \{\lambda_n\}_{n \in \mathbb{N}}$ such that $\lambda_n \rightarrow 0$ and $x \in \lambda_n B$. So then $\forall n \in \mathbb{N}$

$$\frac{x}{\lambda_n} =: x_n \in B,$$

but this is absurd by compactness: as we are working in finite dimension, we get that B is bounded, but if $\|x\|_B \neq 0$, then $\|x_n\|_B \rightarrow \infty$.

We pass on to show that for $a \in \mathbb{R}$, $\|ax\|_B = |a|\|x\|_B$:

$$\begin{aligned}
\|ax\| &= \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid ax \in \lambda B\} \\
&= \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid |a|x \in \lambda B\} \\
&= \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x \in \frac{\lambda}{|a|} B\} \\
&= |a| \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x \in \lambda B\} \\
&= |a| \|x\|_B,
\end{aligned}$$

where the first equality follows from B being symmetric, and the rest from arithmetic manipulations.

We conclude by showing triangle inequality: let $x, y \in \mathbb{R}^n$. By the definition of $\inf$, $\forall \delta > 0$, $\exists \lambda_x, \lambda_y$ such that

$$\begin{aligned}
\lambda_x &\leq \|x\|_B + \delta, & x &\in \lambda_x B \\
\lambda_y &\leq \|y\|_B + \delta, & y &\in \lambda_y B.
\end{aligned}$$

We want to leverage convexity: to do so, we first exclude the case $\lambda_x + \lambda_y = 0$, as then we would have $x = y = 0$. For $\lambda_x + \lambda_y > 0$, we write

$$\frac{x+y}{\lambda_x + \lambda_y} = \frac{\lambda_x}{\lambda_x + \lambda_y} \left(\frac{x}{\lambda_x} \right) + \frac{\lambda_y}{\lambda_x + \lambda_y} \left(\frac{y}{\lambda_y} \right).$$

We get that

$$\frac{x+y}{\lambda_x + \lambda_y} \in B,$$

and then, by the definition of gauge

$$\begin{aligned}
\|x+y\|_B &= \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x+y \in \lambda B\} \\
&\leq \lambda_x + \lambda_y \\
&\leq \|x\|_B + \|y\|_B + 2\delta.
\end{aligned}$$

We conclude by taking $\delta \rightarrow 0$. □

Proposition 1.4. If $E \subset F \subset \mathbb{R}^n$, then $\|x\|_F \leq \|x\|_E \forall x \in \mathbb{R}^n$.

Proof. This follows immediately by the definition:

$$\begin{aligned}
\|x\|_F &= \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x \in \lambda F\} \\
&\leq \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x \in \lambda E\} \\
&= \|x\|_E
\end{aligned}$$

□

1.2 Banach-Mazur distance

In this section we define the Banach-Mazur in two different ways (a technical one and a geometrical one) and we prove their equivalence.

Definition 1.5. (Banach-Mazur Distance) Let E, F be Banach spaces. If E, F are isomorphic, we define the *Banach-Mazur distance* between E and F as

$$d(E, F) := \inf \{ \|T\| \|T^{-1}\| \mid T : E \rightarrow F \text{ isomorphism} \}.$$

If E, F are not isomorphic, then we set $d(E, F) = +\infty$.

The Banach-Mazur distance is a quantity that measures how much the space has to get *linearly distorted* to become another space. The multiplication by the inverse norm grants symmetry and saves us from having to worry about rescaling.

This can be translated in terms of unit balls.

Proposition 1.6. Under the same notations, if we consider B_E, B_F the unit balls of E, F , then

$$d(E, F) = \inf \{ \lambda > 0 \mid \exists T : E \rightarrow F \text{ isomorphism s.t. } T(B_E) \subset B_F \subset \lambda T(B_E) \} \quad (1)$$

Proof. We call the RHS of (1) $\tilde{d}(E, F)$ and prove $d(E, F) = \tilde{d}(E, F)$. We first prove $d(E, F) \leq \tilde{d}(E, F)$. We take $T : E \rightarrow F$ such that $T(B_E) \subset B_F \subset \lambda T(B_E)$ and show $d(E, F) \leq \lambda$. From the inclusions, it follows that

$$\|T\| = \sup_{x \in B_E} \|T(x)\|_F \leq 1,$$

$$\|T^{-1}\| = \sup_{x \in B_F} \|T^{-1}(x)\|_E \leq \lambda,$$

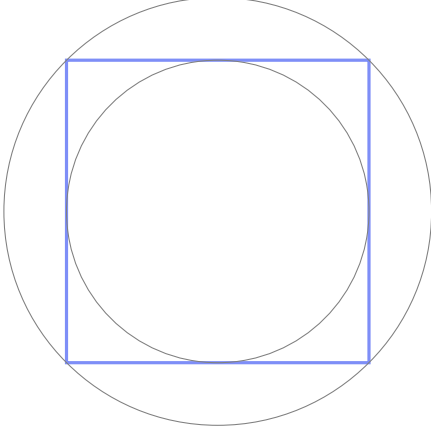
as $T^{-1}(B_F) \subset T^{-1}(T(\lambda B_E)) = \lambda B_E$. This implies that

$$d(E, F) \leq \|T\| \|T^{-1}\| \leq \lambda.$$

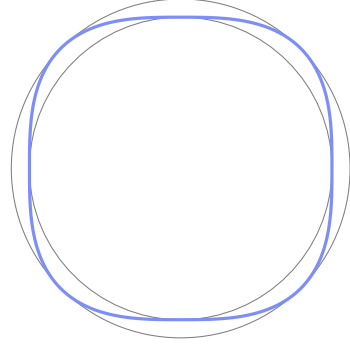
By taking the inf over the T s, we get $d(E, F) \leq \tilde{d}(E, F)$. We move on to prove $d(E, F) \geq \tilde{d}(E, F)$. We take $T : E \rightarrow F$ such that $\|T\| \|T^{-1}\| \leq \lambda$. We can assume $\|T\| = 1$; otherwise, we can rescale. $\|T\| = 1$ implies that $T(B_E) \subset B_F$. Now, as $\|T^{-1}\| = \|T^{-1}\| \|T\| \leq \lambda$, it follows that $T^{-1}(B_F) \subset \lambda B_E$, so that $B_F \subset \lambda T(B_E)$. $\square$

Note: This should give a better picture of what is happening: we are "stretching" the unit ball of E to fit as best as it possibly can into and outside of B_F . We are then measuring the minimal distortion required to identify the two balls. You can expect the distance to be 1 precisely when the two balls coincide, so when E, F are isometric.

Here are a couple of examples:



$$d(\ell^2(\mathbb{R}^2), \ell^\infty(\mathbb{R}^2)) = \sqrt{2}$$



$$d(\ell^2(\mathbb{R}^2), \ell^3(\mathbb{R}^2)) = 2^{1/6}$$

Observation 1.7. For all Banach spaces E, F, G we have

$$d(E, G) \leq d(E, F)d(F, G), \quad (2)$$

as

$$\begin{aligned} \{R \circ S \mid R \in \mathcal{L}(F, G), S \in \mathcal{L}(E, F); R, S \text{ isomorphisms}\} &\subset \\ &\subset \{T \mid T \in \mathcal{L}(E, G), T \text{ isomorphism}\} \end{aligned}$$

implies that

$$\begin{aligned} \inf_{T \in \mathcal{L}(E, G)} \|T\| \|T^{-1}\| &\leq \inf_{\substack{R \in \mathcal{L}(F, G) \\ S \in \mathcal{L}(E, F)}} \|R \circ S\| \|(R \circ S)^{-1}\| \\ &\leq \inf_{\substack{R \in \mathcal{L}(F, G) \\ S \in \mathcal{L}(E, F)}} \|R\| \|S\| \|S^{-1}\| \|R^{-1}\| \\ &\leq \inf_{R \in \mathcal{L}(F, G)} \|R\| \|R^{-1}\| \inf_{S \in \mathcal{L}(E, F)} \|S\| \|S^{-1}\|. \end{aligned}$$

From here on out, we will denote for a generic Banach space E its unit ball by B_E and the identity map on E by I_E . We also denote by $K(n)$ the set of all normed spaces of fixed dimension n .

Proposition 1.8. Two finite dimensional Banach spaces E, F are isometric iff $d(E, F) = 1$.

Proof. We know that $\|T^{-1}\| \geq 1/\|T\|$, so that $\|T\| \|T^{-1}\| \geq 1$ and the inf is reached by the isometry.

Conversely, if $d(E, F) = 1$, we get that $\exists \{\tilde{T}_n\}_{n \in \mathbb{N}}$ such that

$$\|\tilde{T}_n\| \|\tilde{T}_n^{-1}\| \rightarrow 1.$$

We set $T_n := \tilde{T}_n / \|\tilde{T}_n\|$. We get that

$$\|T_n\| \|T_n^{-1}\| = \|\tilde{T}_n\| \|\tilde{T}_n^{-1}\| \rightarrow 1.$$

As $\|T_n\| = 1$, we obtain that $\|T_n^{-1}\| \rightarrow 1$. Now we know E, F are isomorphic finite-dimensional Banach spaces, and so is $\mathcal{L}(E, F)$.

By compactness of, say, $B_0(2)$ we can extract a converging subsequence from the sequences $\{T_n\}_{n \in \mathbb{N}}, \{T_n^{-1}\}_{n \in \mathbb{N}}$ converging to the maps $T : E \rightarrow F, S : F \rightarrow E$, with $\|T\| = \|S\| = 1$. Passing to the limit in $T_n \circ T_n^{-1} = I_F$ and $T_n^{-1} \circ T_n = I_E$, we get that $S = T^{-1}$ and T is an isometry, as desired. $\square$

1.3 John's Lemma

In this subsection, we state (and do not prove) a fundamental result published by John in 1948. John's theorem is a result telling us a lot on how n -dimensional Banach spaces interact, and gives us a bound on the Banach-Mazur distance, as well as a uniform bound on the norms in term of the euclidean norm.

Definition 1.9. Let E be an n dimensional Banach space. Then $D \subset E$ is an *ellipsoid* if there exists an isomorphism $T : \ell_2^n \rightarrow E$ such that $T(B_{\ell_2^n}) = D$

Theorem 1.10. (John's Lemma) Let E be an n -dimensional Banach space. Then there exists a unique ellipsoid D_E of maximal volume contained into B_E . Moreover

$$B_E \subset \sqrt{n}D_E.$$

By definition of Banach-Mazur distance, we get the following corollary:

Corollary 1.11. Let E, F be n -dimensional Banach spaces. Then $d(E, \ell_2^n) \leq \sqrt{n}$, and it also follows that $d(E, F) \leq n$.

The first part follows immediately by the definition, the second follows from (2). Note that the estimate is sharp, as $d(\ell_2^n, \ell_\infty^n) = \sqrt{n}$ (see the examples in the previous section, with $n = 2$).

Note: These corollaries, as well as proposition 1.8 motivate us to check the properties of a space including all n -dimensional Banach spaces, and this will be our next step.

1.4 Banach-Mazur Compactum

In this subsection, we define the Banach-Mazur compactum: this will become our ambient space for working on Dvoretzky's theorem, and we will soon understand why.

Theorem 1.12. Let us denote by $\tilde{K}(n)$ the set of classes of $K(n)$ modulo isometry. Then $\tilde{K}(n)$ equipped with the metric $\delta(E, F) = \log d(E, F)$ is a compact metric space.

Proof. The fact that δ is a distance on $\tilde{K}(n)$ follows from (2), proposition 1.8, and symmetry of Banach-Mazur distance. We want to leverage the quotient to choose a suitable representative for every space in $\tilde{K}(n)$, which will be $\mathbb{R}^n$ with a suitable norm. We use John's theorem, written in the following way: we know there exists a linear isomorphism $T : \mathbb{R}^n \rightarrow E$ such that

$$T(B_{\ell_2^n}) \subset B_E \subset \sqrt{n}T(B_{\ell_2^n}),$$

which implies

$$B_{\ell_2^n} \subset T^{-1}(B_E) \subset \sqrt{n}B_{\ell_2^n},$$

We consider the following space:

$$(\mathbb{R}^n, \|\cdot\|_{T^{-1}(B_E)}).$$

This is isometric to $(E, \|\cdot\|_E)$ as

$$\begin{aligned} \|x\|_{T^{-1}(B_E)} &= \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid x \in \lambda T^{-1}(B_E)\} \\ &= \inf\{\lambda \in \mathbb{R}_{\geq 0} \mid Tx \in \lambda B_E\} \\ &= \|Tx\|_{B_E}. \end{aligned}$$

Note that, by proposition 1.4, we get that

$$\frac{1}{\sqrt{n}}\|x\|_{\ell_2^n} \leq \|x\|_{T^{-1}(B_E)} \leq \|x\|_{\ell_2^n} \quad \forall x \in \mathbb{R}^n, \quad (3)$$

which gives

$$\frac{1}{\sqrt{n}} \leq \|x\|_{T^{-1}(B_E)} \leq 1 \quad \forall x \in \mathbb{S}^{n-1}, \quad (4)$$

Now that we have moved to $\mathbb{R}^n$ we can prove compactness. Take $\{(\mathbb{R}^n, \|\cdot\|_k)\}_{k \in \mathbb{N}} \subset \tilde{K}(n)$. The norms are all uniquely defined by their value on $\mathbb{S}^{n-1}$ (by dividing and multiplying for $\|x\|_{\ell_2^n}$ we can always bring ourselves back to $\mathbb{S}^{n-1}$), which is compact. We use Ascoli-Arzelà: the bound in (4) gives us equiboundedness, while equicontinuity follows from (3). We get a uniform limit $\|\cdot\|_\infty$, and the limit element in $\tilde{K}(n)$ will be $(\mathbb{R}^n, \|\cdot\|_\infty)$. $\square$

Note: The uniform bound on the norm is possible only because of the ambient space: else the bound with the ℓ_2^n norm wouldn't be possible: it would be enough to take a subset of $B_{\ell_2^n}$ and the gauge would be bigger. The Banach-Mazur compactum lets us do just that: picking a representative where the unit ball is "rescaled" in some sense, letting us work with uniform bounds that we wouldn't otherwise have.

Definition 1.13. We call the space $(\tilde{K}(n), \delta)$ the *Banach-Mazur compactum*.

1.5 Duals

We now use the correspondence between balls and norms to define the dual (we will call it *polar*) of a ball. We will use this to dualize Dvoretzky's Theorem in Chapter 2.

Definition 1.14. Let B be a ball, $E = (\mathbb{R}^n, \|\cdot\|_B)$. Then we call *polar* of B , denoted as B° , the unit ball of E^* .

Proposition 1.15. Maintaining the notations above, by using the scalar product to identify E and E^* , we can characterize the polar as

$$B^\circ = \{y \in \mathbb{R}^n : x \cdot y \leq 1 \quad \forall x \in B\}.$$

Proof. The proof consists in noticing two things: first that for any linear function $T : E \rightarrow \mathbb{R}$ we have that $Tx = y \cdot x$ for some $y \in \mathbb{R}^n$ and foremost that

$$\|T\|_{E^*} = \sup_{\|x\| \leq 1} |y \cdot x| = \sup_{x \in B} |y \cdot x|$$

So it follows

$$B^\circ = \{T : E \rightarrow \mathbb{R} : \|T\|_{E^*} \leq 1\} = \{y \in \mathbb{R}^n : x \cdot y \leq 1 \ \forall x \in B\}, \quad (5)$$

where the equality is up to identification. $\square$

Proposition 1.16. If $E \subset \mathbb{R}^n$ is a linear subspace, then

$$(B \cap E)^\circ = P_E(B^\circ),$$

where P_E is the projection of $\mathbb{R}^n$ onto E , and the polar is taken with respect to E .

Proof. We first prove $\supseteq$: let $x \in P_E(B^\circ)$, we want to show that $x \in (B \cap E)^\circ$, which means $x \cdot y \leq 1 \ \forall y \in B \cap E$. We know that $x = P_E(z)$ for some $z \in B^\circ$. Then, $z \cdot y \leq 1 \ \forall y \in B$. This means that $\forall y \in B \cap E$

$$x \cdot y = P_E(z) \cdot y = z \cdot y - (z - P_E(z)) \cdot y = z \cdot y \leq 1,$$

by which we conclude. We now prove the inclusion $\subseteq$: let $x \in (B \cap E)^\circ$, we want to find $z \in B^\circ$ such that $x = P_E(z)$. We consider $f : E \rightarrow \mathbb{R}$ defined as $f(y) = x \cdot y$. By hypothesis, $\|f\|_\infty \leq 1$, or equivalently $|f(y)| \leq \|y\|_B$. By Hahn-Banach we can extend f to a linear $F : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $\|F\|_\infty \leq 1$. We conclude by Riesz Representation Theorem, as it implies that $\exists z \in \mathbb{R}^n$ such that $z \cdot y \leq 1 \ \forall y \in B$ and $P_E(z) = x$. $\square$

1.6 Orthonormal Systems

Now that we are done with defining our spaces and distances, we are interested in Theorems that help us bring the geometry forth from one space to another. This is what we are going to do in the next two sections.

In particular, in this section we are going to state and prove a Lemma that will help us move orthonormal systems from one space to the other using linear isomorphisms. We now define the α_n coefficients which will help us do just that.

Definition 1.17. Let $T : X \rightarrow Y$ be an operator between Banach spaces. We denote as

$$\alpha_n(T) = \inf\{\|T - S\| \mid S : X \rightarrow Y \text{ rk}(S) < n\}.$$

Note: The α_n -s are estimating how close T is to an n -rank map. if the α_n s are zero from some point on, T is a finite rank map. Note that $\|T\| = \alpha_1(T) \geq \alpha_2(T) \geq \dots$

Lemma 1.18. Let $T : H \rightarrow X$ be an operator from a Hilbert space into a Banach space. For every $\epsilon > 0$, there is an orthonormal system $(f_n)_{n \in \mathbb{N}}$ in H such that $\|T(f_n)\| \geq \alpha_n(T) - \epsilon$ for all $n \geq 1$. Moreover if H is finite dimensional, we have this for also $\epsilon = 0$

Proof. Let $f_1 \in H$ such that $\|f_1\| = 1$ and

$$\|T(f_1)\| \geq \|T\| - \epsilon = \alpha_1 - \epsilon.$$

We consider the restriction of T to $S_1 = \langle f_1 \rangle^\perp$ and note that $\|T \upharpoonright_{S_1}\| \geq \alpha_2(T)$, since if we denote P the projection onto $\langle f_1 \rangle$, $\text{rk}(T \circ P) \leq 1$ and hence $\alpha_2 \leq \|T - T \circ P\| = \|T \upharpoonright_{S_1}\|$. It follows there is a norm 1 element $f_2 \in \langle f_1 \rangle^\perp$ such that

$$\|T(f_2)\| \geq \alpha_2(T) - \epsilon.$$

We then restrict T to $\langle f_1, f_2 \rangle^\perp$. Continuing in this way, we obtain a sequence $(f_n)_{n \in \mathbb{N}}$ with the desired property. In the finite dimensional case, since the unit balls are compact, we can reduce to $\epsilon = 0$. $\square$

Note: This lower bound will be fundamental to ensure that when we move specific orthonormal systems, they do not end up in zero.

We really need these kinds of rank approximation: we are going to be working with spaces for which we want to see how the behavior changes when the dimension rises, and we need to treat the maps between the spaces accordingly as if we were already in infinite dimension.

1.7 Lewis' Theorem

In this section we talk about Lewis' Theorem, which will help us find particular isomorphisms between $\mathbb{R}^n$ and Banach spaces. This, combined with Lemma 1.17 will give us the tools to build sets of vectors resembling orthonormal bases in the next chapter (see Lemma 2.16).

Definition 1.19. Let E be a Banach space, α a norm on the space $\mathcal{L}(\mathbb{R}^n, E)$ of all linear operators from $\mathbb{R}^n$ into E . We introduce on $\mathcal{L}(\mathbb{R}^n, E)$ a dual norm α^* as follows:

$$\forall v : E \rightarrow \mathbb{R}^n \quad \alpha^*(v) = \sup\{\text{tr}(vT) \mid T : \mathbb{R}^n \rightarrow E \text{ s.t. } \alpha(T) \leq 1\}. \quad (6)$$

We now state Lewis' Theorem.

Theorem 1.20 (Lewis' Theorem). Let E be a vector space of dimension n and let α be a norm on $\mathcal{L}(\mathbb{R}^n, E)$. Then there is an isomorphism $u : \mathbb{R}^n \rightarrow E$ such that

$$\alpha(u) = 1 \text{ and } \alpha^*(u^{-1}) = n.$$

Proof. Let $u \rightarrow \det(u)$ be a determinant function (associated to a fixed linear basis of E). Let $K = \{u \in \mathcal{L}(\mathbb{R}^n, E) \mid \alpha(u) \leq 1\}$. Since K is compact, by Wierstrass' Theorem, there exists u in K such that

$$|\det(u)| = \sup\{|\det(v)| \mid v \in K\}.$$

This implies that $\forall T \in \mathcal{L}(\mathbb{R}^n, E)$,

$$\left| \det\left(\frac{u+T}{\alpha(u+T)}\right) \right| \leq |\det(u)|,$$

hence by homogeneity,

$$|\det(u+T)| \leq |\det(u)|(\alpha(u+T))^n. \quad (7)$$

Being u an isomorphism, $\det(u) \neq 0$, dividing (7) by $\det(u)$, we find

$$\begin{aligned} |\det(1 + u^{-1}T)| &\leq \alpha(u+T)^n \\ &\leq (1 + \alpha(T))^n \end{aligned}$$

by triangle inequality. Since T is arbitrary, we have $\forall \epsilon > 0$

$$|\det(1 + \epsilon u^{-1}T)| \leq (1 + \epsilon \alpha(T))^n. \quad (8)$$

When $\epsilon \rightarrow 0$, $\det(1 + \epsilon u^{-1}T) = 1 + \epsilon \text{tr}(u^{-1}T) + o(\epsilon)$; hence (8) implies

$$\text{tr}(u^{-1}T) \leq n\alpha(T)$$

for all $T : E \rightarrow \mathbb{R}^n$. Equivalently, by (6)

$$\alpha^*(u^{-1}) \leq n.$$

On the other hand, we have $n = \text{tr}(u^{-1}u) \leq \alpha(u)\alpha^*(u^{-1})$, hence $\alpha(u) = 1$ and $\alpha(u^{-1}) = n$. $\square$

1.8 Gaussian Measures

In this last section we recall the fundamental properties of Gaussian measures. We will denote as $(\Omega, \mathcal{A}, \mathbb{P})$ a suitable probability space and by $(g_k)_{k \in \mathbb{N}}$ a sequence of independent Gaussian random variables with the standard normal distribution, γ_n the standard Gaussian probability measure on $\mathbb{R}^n$, so that for $B \subset \mathbb{R}^n$ Borel, we have

$$\begin{aligned} \gamma_n &= (2\pi)^{-n/2} \exp\left(-\frac{1}{2} \sum_{i=1}^n x_i^2\right) dx_1 \dots dx_n, \\ \mathbb{P}(\{(g_1, \dots, g_n) \in B\}) &= (2\pi)^{-n/2} \int_B \exp\left(-\frac{1}{2} \sum_{i=1}^n x_i^2\right) dx_1 \dots dx_n = \gamma_n(B). \end{aligned}$$

The property which will be of crucial importance in the sequel is the rotational invariance of the Gaussian distribution, i.e. if (a_{ij}) is an orthogonal $n \times n$ matrix, then the sequence

$$\tilde{g}_i = \sum_{j=1}^n a_{ij} g_j, \quad i = 1, \dots, n$$

has the same distribution on $\mathbb{R}^n$ as the original sequence $(g_1, \dots, g_n)$, which means that for every $B \subset \mathbb{R}^n$ Borel,

$$\mathbb{P}(\{(\tilde{g}_1, \dots, \tilde{g}_n) \in B\}) = \mathbb{P}(\{(g_1, \dots, g_n) \in B\}). \quad (9)$$

This follows immediately from the invariance under $O(n)$ of the standard Gaussian probability measure on $\mathbb{R}^n$. Let now X be a random variable with values in a separable Banach space E . We will denote by $\text{dist}(X)$ the distribution of X . Then (9) has many consequences: for any $\alpha \in \mathbb{R}^n$, we have

$$\text{dist}\left(\sum_{i=1}^n \alpha_i g_i\right) = \text{dist}\left(g_1 \left(\sum_{i=1}^n |\alpha_i|^2\right)^{1/2}\right). \quad (10)$$

Indeed, by homogeneity we can assume $\|\alpha\|_{\ell_2^n} = 1$, and then just use the fact that there is an A in $O(n)$ mapping α into e_1 . By (10) we have

$$\left\| \sum_{i=1}^n \alpha_i g_i \right\|_p = \|g_1\|_p \left(\sum_{i=1}^n |\alpha_i|^2 \right)^{1/2}. \quad (11)$$

We also recall that

$$\mathbb{E}|g_1| = \sqrt{2/\pi}, \quad (12)$$

and more in general

$$\begin{aligned} \|g_1\|_q &= \left(\frac{2}{\sqrt{2\pi}} \int_0^\infty x^q e^{-x^2/2} dx \right)^{1/q} \\ &\leq K \sqrt{q}, \end{aligned} \quad (13)$$

where K is a numerical constant. After recalling these facts, we now prove three Lemmas that will help us deal with calculations in the proof for Dvoretzky's Theorem.

Lemma 1.21. For any $\zeta \in \mathbb{R}^n$, we have

$$\int_{\mathbb{R}^n} \exp(\zeta \cdot y) d\gamma_n(y) = \exp \frac{1}{2} \|\zeta\|_{\ell_2^n}^2$$

Proof. We just need to prove this for $n = 1$, the rest will follow, as by Fubini-Tonelli

$$\int_{\mathbb{R}^n} \exp(\zeta \cdot y) d\gamma_n(y) = \prod_{i=1}^n \int_{\mathbb{R}} \exp(\zeta_i y_i) d\gamma_1(y_i).$$

For the case $\zeta \in \mathbb{R}$, we conclude by observing

$$\begin{aligned} \int_{\mathbb{R}} \exp(\zeta y) d\gamma_1(y) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(\zeta y - \frac{y^2}{2}\right) dy \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(\frac{\zeta^2}{2} - \frac{(y - \zeta)^2}{2}\right) dy \\ &= \exp\left(\frac{\zeta^2}{2}\right) \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{(y - \zeta)^2}{2}\right) dy\right) \\ &= \exp\left(\frac{\zeta^2}{2}\right). \end{aligned}$$

□

Lemma 1.22. Let E be a Banach space, $n \in \mathbb{N}$, X_i , $i \leq n$ random variables defined as follows:

$$X_i = \sum_{j=1}^{\infty} g_j^i z_j,$$

where $(g_j^i)_{i,j \in \mathbb{N}}$ are independent standard Gaussian random variables, and $(z_j)_{j \in \mathbb{N}} \subset E$. Suppose X_i converges in $L_1(\Omega, \mathcal{A}, \mathbb{P})$ for all $i \leq n$. Then, for all $\alpha \in \mathbb{R}^n$ such that $\|\alpha\|_{\ell_2^n} = 1$, we get

$$\text{dist}\left(\sum_{i=1}^n \alpha_i X_i\right) = \text{dist}(X_1).$$

Proof. We simply write down the variable:

$$\begin{aligned} \sum_{i=1}^n \alpha_i X_i &= \sum_{i=1}^n \alpha_i \left(\sum_{j=1}^{\infty} g_j^i z_j \right) \\ &= \sum_{j=1}^{\infty} \left(\sum_{i=1}^n \alpha_i g_j^i \right) z_j. \end{aligned}$$

We use the fact that the linear combination of normal Gaussians is still a normal Gaussian with variance

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n \alpha_i g_j^i\right) &= \sum_{i=1}^n \alpha_i^2 \text{Var}(g_j^i) \\ &= \sum_{i=1}^n \alpha_i^2 = 1 \end{aligned}$$

and we conclude, as

$$\text{dist}\left(\sum_{i=1}^n \alpha_i g_j^i\right) = \text{dist}(g_j^1).$$

□

Lemma 1.23. Let g be a standard Gaussian measure. Then $\exists \alpha > 0$ such that for all $N > 1$, $N \in \mathbb{N}$, we get

$$\mathbb{P}\left(\left\{g > \alpha(\log N)^{1/2}\right\}\right) \geq 1/N$$

Proof. We first prove that for any $x > 0$,

$$\mathbb{P}\left(\left\{g > x\right\}\right) \geq \frac{x}{\sqrt{2\pi}(x^2 + 1)} \exp\left(-\frac{x^2}{2}\right) \quad (14)$$

Let $f(t) = -1/t \exp(-t^2/2)$. Then for all $t > x$, we have

$$\begin{aligned} f'(t) &= \left(\frac{1}{t^2} + 1\right) \exp\left(-\frac{t^2}{2}\right) \\ &\leq \frac{x^2 + 1}{x^2} \exp\left(-\frac{t^2}{2}\right), \end{aligned}$$

by which we get

$$\begin{aligned} \frac{1}{x} \exp\left(-\frac{x^2}{2}\right) &= \int_x^\infty f'(t) dt \\ &\leq \frac{x^2 + 1}{x^2} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt. \end{aligned}$$

By this, (14) follows, as

$$\begin{aligned} \mathbb{P}\left(\left\{g > x\right\}\right) &= \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt \\ &\geq \frac{x}{\sqrt{2\pi}(x^2 + 1)} \exp\left(-\frac{x^2}{2}\right). \end{aligned}$$

We just need to substitute $x = \alpha(\log N)^{1/2}$ and choose the right α : we get

$$\exp\left(-\frac{x^2}{2}\right) = \exp\left(-\frac{\alpha^2 \log N}{2}\right) = N^{-\alpha^2/2};$$

substituting in (14) we get

$$\mathbb{P}\left(\left\{g > \alpha(\log N)^{1/2}\right\}\right) \geq \frac{\alpha(\log N)^{1/2}}{\sqrt{2\pi}(\alpha^2 \log N + 1)} N^{-\alpha^2/2},$$

which for any $\alpha < \sqrt{2}$ gives the thesis. □

Chapter 2

Dvoretzky's Theorem

We now state Dvoretzky's Theorem. The original proof appeared in [1]. For subsequent proofs, see [3], [6] and [2]. In this chapter we follow the proof from [5], which is the measure-theoretical point of view of Milman's proof ([3]), readapted by Pisier with the use of Gaussian measures instead of surface measures on Euclidean spheres.

Theorem 2.1. (Dvoretzky's Theorem) For each $\epsilon > 0$ and $n \in \mathbb{N}$, there is a number $N(\epsilon, n)$ such that every Banach space E of dimension at least $N(\epsilon, n)$ contains an n -dimensional subspace F such that $d(F, \ell_2^n) \leq 1 + \epsilon$.

We notice the following consequence for infinite-dimensional Banach spaces.

Corollary 2.2. Let E be an infinite-dimensional Banach space. Then, for each $\epsilon > 0$ there is a sequence $\{E_n\}_{n \in \mathbb{N}}$ of subspaces of E such that

$$d(E_n, \ell_2^n) \leq 1 + \epsilon.$$

We can always reduce ourselves to the finite-dimensional case, so this is the one we focus on. We prove the following refinement of Theorem 2.1:

Theorem 2.3. For each $\epsilon > 0$, there is a number $\eta(\epsilon) > 0$ with the following property. Let E be a finite dimensional Banach space of dimension N . Then E contains a subspace $F \subset E$ of dimension

$$n = \lfloor \eta(\epsilon) \log N \rfloor$$

such that

$$d(F, \ell_2^n) \leq 1 + \epsilon.$$

Remark 2.4. By the equivalent definition of Banach-Mazur distance, the above theorem says that if

$$n = \lfloor \eta(\epsilon) \log N \rfloor,$$

then any ball $B \subset \mathbb{R}^N$ admits a section which is $(1 + \epsilon)$ -equivalent to an ellipsoid, i.e. there is a subspace $F \subset \mathbb{R}^N$ with $\dim F = n$ such that

$$D \subset F \cap B \subset (1 + \epsilon)D$$

By duality, a similar statement holds for projections of B : there is an n -dimensional projection of B which is $(1 + \epsilon)$ -equivalent to an ellipsoid. In other words, every normed space E of dimension N admits, for $n = \lfloor \eta(\epsilon) \log N \rfloor$ an n -dimensional quotient space which is $(1 + \epsilon)$ -isomorphic to ℓ_2^n .

Remark 2.5. Theorem 2.3 implies theorem 2.1 with

$$N(\epsilon, n) = \exp \frac{n}{\eta(\epsilon)} \tag{15}$$

Theorem 2.6. For each $\epsilon > 0$ there is a number η_1 such that the following holds. Let $(g_k)_{k \in \mathbb{N}}$ i.i.d. sequence of Gaussian normal random variables on some probability space $(\Omega, \mathcal{A}, \mathcal{P})$. Let E be a Banach space, and let $(z_k)_{k \in \mathbb{N}}$ be a sequence of elements of E . We assume that the series

$$X = \sum_{k \in \mathbb{N}} g_k z_k \quad (16)$$

converges in $L_1(\Omega, \mathcal{A}, \mathbb{P})$. Let

$$\sigma(X) = \sup \left\{ \left(\sum_{k \in \mathbb{N}} |\zeta(z_k)|^2 \right)^{1/2} \mid \zeta \in E^* : \|\zeta\| \leq 1 \right\},$$

and

$$d(X) = \left(\frac{\mathbb{E}\|X\|}{\sigma(X)} \right)^2.$$

Then, for each $\epsilon > 0$, E contains a subspace F of dimension

$$n = \lfloor \eta_1(\epsilon) d(X) \rfloor$$

which is $(1 + \epsilon)$ -isomorphic to ℓ_2^n .

Note: The point of taking these types of X -s is to choose a sequence of vectors in E and taking vectors randomly in the span. We use Gaussians because we want the coefficients to resemble lengths and Gaussians have spherical symmetry. Moreover they will come in handy to leverage the symmetry of the problem.

The scalar $\sigma(X)$ measures *how perpendicular* the vectors in the sequence are: try to think of this in $\mathbb{R}^n$. Then ζ is a projection onto a unit vector. Then $\sigma(X)$ is maximal whenever the z_k are all parallel, for ζ equal to their direction. So σ gets bigger and bigger when we concentrate our vectors on the same direction, so that it represents the reciprocal of some kind of notion of dimension (more like concentration of dimension). We multiply by $\mathbb{E}(\|X\|)$ to make it invariant under multiplications by constants: in this way if the z_k -s are all parallel, $d(X) = 1$. We also square because of the following remark.

Remark 2.7. We have that, by (10) and (12) $\mathbb{E}|\zeta(X)| = \mathbb{E}|g_1|(\sum |\zeta(z_k)|^2)^{1/2}$ and $\mathbb{E}|g_1| = (2/\pi)^{1/2}$. Hence

$$\sigma(X) \leq \left(\frac{\pi}{2} \right)^{1/2} \mathbb{E}\|X\|.$$

It follows that

$$d(X) \geq \frac{2}{\pi}.$$

which ensures that $d(X)$ does not collapse. Moreover, if E is finite-dimensional ($\dim(E) = N$), then we know that

$$\mathbb{E}\|X\| \leq (\mathbb{E}\|X\|^2)^{1/2} \leq N^{1/2} \sigma(X),$$

which, again, implies

$$d(X) \leq N.$$

In the next pages we will work more towards checking the behaviour of $d(X)$ (in particular we will be interested in the quantity $\delta(E)$ defined as follows.

Remark 2.8. Let $\delta(E) = \sup(\delta(X))$ where the supremum runs all over E -valued Gaussian variables X as in theorem 2.6, and let n_ϵ be the largest integer such that there is an n -dimensional subspace $F \subset E$ satisfying $d(F, \ell_2^n) \leq 1 + \epsilon$. By Theorem 2.6 we have for some function $\eta_1(\epsilon) > 0$

$$n_\epsilon(E) \geq \eta_1(\epsilon)\delta(E).$$

There is also a converse estimate which shows that this measure theoretic approach is somewhat sharp, and it is much easier to check as follows.

Proposition 2.9. Maintaining the notations of the previous remark, we have

$$\frac{\pi}{2}\delta(E)(1 + \epsilon)^2 \geq n_\epsilon(E)$$

Proof. Let $F \subset E$ be such that $d(F, \ell_2^n) \leq 1 + \epsilon$. Let $T : \ell_2^n \rightarrow F$ be such that $\|T\|\|T^{-1}\| \leq 1 + \epsilon$. We define

$$X = \sum_{i=1}^n g_i T(e_i)$$

Then $\sigma(X) = \|T\|$, as for $\zeta \in E^*$ we have that $\zeta \circ T \in (\ell_2^n)^*$ linear and continuous, so that it exists an element $x \in \ell_2^n$ such that $\forall y \in \ell_2^n$

$$\zeta \circ T(y) = \langle y, x \rangle,$$

and automatically,

$$\begin{aligned} \|\zeta \circ T\| &= \|x\|_{\ell_2^n} \\ &= \left(\sum_{i=1}^n (x_i)^2 \right)^{1/2} \\ &= \left(\sum_{i=1}^n (\zeta \circ T(e_i))^2 \right)^{1/2}; \end{aligned}$$

so by taking the sup over ζ -s we get $\sigma(X) = \|T\|$, as we wanted. Then,

$$\begin{aligned} \|T^{-1}\| \mathbb{E}\|X\| &\geq \mathbb{E}\|T^{-1}(X)\| \\ &= \mathbb{E} \left(\sum_{i=1}^n |g_i|^2 \right)^{1/2} \\ &\geq \left(\sum_{i=1}^n (\mathbb{E}|g_i|)^2 \right)^{1/2} \\ &= n^{1/2} \left(\frac{2}{\pi} \right)^{1/2}. \end{aligned}$$

This implies $d(X) \geq (2/\pi)n(\|T\|\|T^{-1}\|)^{-2} \geq (2/\pi)n(1 + \epsilon)^{-2}$, yielding the proposition. $\square$

Note: We are leveraging the fact that T is almost an isomorphism to find reasonably perpendicular vectors, and maximize X as much as we can.

Strategy of proof Theorem 2.6 is proved in detail below, but we want to give first a general idea of the proof. Let $X_1, X_2, \dots, X_n$ be i.i.d. copies of the variable X on some probability space

$(\Omega, \mathcal{A}, \mathbb{P})$. Let $\Omega(n, \epsilon)$ be the set of all $\omega \in \Omega$ such that $\forall (\alpha_i)_{i \in \mathbb{N}} \subset \mathbb{R}^n$

$$(1 + \epsilon)^{-1/2} \left(\sum_{i=1}^{\infty} |\alpha_i|^2 \right)^{1/2} \leq \left\| \sum_{i=1}^{\infty} \alpha_i X_i(\omega) (\mathbb{E} \|X\|)^{-1} \right\| \leq (1 + \epsilon)^{1/2} \left(\sum_{i=1}^{\infty} |\alpha_i|^2 \right)^{1/2}. \quad (17)$$

We will show that $\mathbb{P}(\Omega(n, \epsilon)) > 0$, provided that $n \leq \lfloor \eta_2(\epsilon) d(X) \rfloor$. This yields theorem 2.6.

Note: We now work on some theorems and lemmas necessary to the actual proof of theorem 2.6. We begin with a concentration of measure result which will be crucial.

Theorem 2.10. Let X be as in Theorem 2.6. We have

$$\mathbb{P}\{\|X\| - \mathbb{E}\|X\| > t\} \leq 2 \exp(-Kt^2/\sigma(X)^2) \quad \forall t > 0, \quad (18)$$

where K is a numerical constant ($K = 2\pi^{-2}$ in the proof below)

Proof. We can assume E is finite dimensional. In this way, the series (16) is a finite sum,

$$X = \sum_{k=1}^m g_k z_k \quad (z_k \in E).$$

We can also assume non-degeneracy, so that the distribution of X is equivalent to the Lebesgue measure on E (else we just pass to a subspace). Let $u : \ell_2^m \rightarrow E$ be the linear operator defined by

$$u(x) = \sum x_k z_k \quad \forall x \in \mathbb{R}^m,$$

and γ_m the canonical Gaussian probability measure on $\mathbb{R}^m$. Then (18) reduces to the following inequality:

$$\gamma_m\left(\left\{x \in \mathbb{R}^m \mid \left| \|u(x)\| - \int \|u(x)\| d\gamma_m \right| > t \right\}\right) \leq 2 \exp\left(-\frac{Kt^2}{\|u\|^2}\right). \quad (19)$$

Indeed, $\sigma(X) = \|u^*\| = \|u\|$, where u^* is the adjoint operator of u , as

$$\begin{aligned} \|u\| &= \|u^*\| = \sup_{\|\zeta\| \leq 1} \|\zeta(u)\|_{\ell_2^{n*}} \\ &= \sup_{\|\zeta\| \leq 1} \sup_{\|x\|_{\ell_2^m} = 1} \left| \sum_{k=1}^m x_k \zeta(z_k) \right| \\ &= \sup_{\|\zeta\| \leq 1} \left(\sum_{k=1}^m |\zeta(z_k)|^2 \right)^{1/2} \\ &= \sigma(X). \end{aligned}$$

To prove (19) we will work on $\mathbb{R}^m \times \mathbb{R}^m$ to leverage the invariance by rotation of the Gaussian measures. We will denote by (x, y) a generic point of $\mathbb{R}^m \times \mathbb{R}^m$. For $\theta \in [0, 2\pi]$, let then

$$\begin{aligned} x(\theta) &= x \sin \theta + y \cos \theta, \\ x'(\theta) &= x \cos \theta - y \sin \theta; \end{aligned}$$

so that we are rotating clockwise of an angle θ and then exchanging the axes (x' is the rotated x axis, and x the rotated y axis). We also denote

$$F(x) = \|u(x)\|.$$

Note that F is Lipschitz: u is linear and continuous so we get

$$\begin{aligned} |F(x) - F(y)| &= \left| \|u(x)\| - \|u(y)\| \right| \\ &\leq \|u(x - y)\| \\ &\leq \|u\| \|x - y\|_{\ell_2^n}. \end{aligned}$$

Notice that this implies that by Rademacher's theorem, F admits a derivative at almost every point $x \in \mathbb{R}^m$, and moreover

$$\|F'(x)\|_{\ell_2^n} \leq \|u\|. \quad (20)$$

Now let x, y be fixed and assume now that F is differentiable in $x(\theta)$ for almost every θ . Then we can write

$$\begin{aligned} F(x) - F(y) &= F(x(\pi/2)) - F(x(0)) \\ &= \int_0^{\pi/2} \frac{d}{d\theta} F(x(\theta)) d\theta \\ &= \int_0^{\pi/2} F'(x(\theta)) x'(\theta) d\theta. \end{aligned} \quad (21)$$

Let λ be a real number and let $\Phi_\lambda : \mathbb{R} \rightarrow \mathbb{R}$ be the function $\Phi_\lambda(t) = \exp(\lambda t)$ for all t in $\mathbb{R}$. By convexity, we deduce from (21)

$$\begin{aligned} \Phi_\lambda(F(x) - F(y)) &= \Phi_\lambda\left(\int_0^{\pi/2} F'(x(\theta)) x'(\theta) d\theta\right) \\ &\leq \frac{2}{\pi} \int_0^{\pi/2} \Phi_\lambda\left(F'(x(\theta)) x'(\theta) \frac{\pi}{2}\right) d\theta. \end{aligned} \quad (22)$$

We now wish to integrate (22) in x, y with respect to $\gamma_m \times \gamma_m$. Note that since $F'(x)$ exists for a.e. x , for a.e. couple (x, y) the derivative $F'(x(\theta))$ exists for a.e. θ , as:

$$\gamma_m \times \gamma_m(\{(x, y) \in \mathbb{R}^m \times \mathbb{R}^m \mid \nexists F'(x(\theta))\}) = \gamma_m \times \gamma_m(\{(x, y) \in \mathbb{R}^m \times \mathbb{R}^m \mid \nexists F'(x)\})$$

by invariance by rotations once more; $F(x(\theta))$ will be differentiable for a.e. (x, y) since the measure of the set above is 0. It also follows $F'(x(\theta))$ exists for a.e. θ by Fubini's theorem (integrating three times the set of non-differentiability). We obtain

$$\iint \Phi_\lambda(F(x) - F(y)) d\gamma_m(x) d\gamma_m(y) \leq \frac{2}{\pi} \iint \int_0^{\pi/2} \Phi_\lambda\left(\frac{\pi}{2} F'(x(\theta)) x'(\theta)\right) d\theta d\gamma_m(x) d\gamma_m(y). \quad (23)$$

By convexity we get

$$\int \Phi_\lambda\left(F(x) - \int F(y) d\gamma_m(y)\right) d\gamma_m(x) \leq \iint \Phi_\lambda(F(x) - F(y)) d\gamma_m(x) d\gamma_m(y).$$

We now finally leverage the invariance by rotation of the Gaussian measures: for every θ , the measure $\gamma_m \times \gamma_m$ is invariant under the mapping $(x, y) \rightarrow (x(\theta), x'(\theta))$. Therefore the right-hand side of (23) is equal to

$$\iint \Phi_\lambda\left(\frac{\pi}{2} F'(x) \cdot y\right) d\gamma_m(x) d\gamma_m(y)$$

by using Fubini's theorem again. By Lemma 1.21, we know that for any $\zeta \in \mathbb{R}^m$

$$\int \exp(\zeta(y)) d\gamma_m(y) = \exp\left(\frac{1}{2} \|\zeta\|_{\ell_2^n}^2\right),$$

hence

$$\int \exp\left(\frac{\pi}{2}\lambda F'(x)y\right)d\gamma(y) = \exp\frac{1}{2}\left(\frac{\pi}{2}\right)^2 \|F'(x)\|_{\ell_2^n}^2 \lambda^2;$$

but by (20), together with (23) gives that $\forall \lambda \in \mathbb{R}$

$$\int \Phi_\lambda\left(F(x) - \int F(y)d\gamma_m(y)\right)d\gamma_m(x) \leq \exp\frac{1}{2}\left(\frac{\pi}{2}\right)^2 \|u\|^2 \lambda^2.$$

By Markov inequality we have that $\forall \lambda > 0$

$$\begin{aligned} \gamma_m\left(\left\{x \mid F(x) - \int F d\gamma_m > t\right\}\right) &\leq \frac{\mathbb{E}\left[\exp\lambda\left(F(x) - \int F d\gamma_m\right)\right]}{\exp(\lambda t)} \\ &\leq \exp\left(\frac{\pi^2}{8}\|u\|^2 \lambda^2 - \lambda t\right). \end{aligned}$$

Choosing $\lambda = (\pi\|u\|/2)^{-2}t$, we find

$$\gamma_m\left(\left\{x \mid F(x) - \int F d\gamma_m > t\right\}\right) \leq \exp -\frac{1}{2}\left(\frac{\pi}{2}\|u\|\right)^{-2} t^2.$$

The same inequality holds for $-F$ also, so that we finally obtain the announced inequality (19) and hence also (18). $\square$

Remark 2.11. Let's see how we can reinterpret these results for different functions and measures: the preceding proof actually establishes the following: for any Lipschitzian function $F : \mathbb{R}^m \rightarrow \mathbb{R}$ such that

$$\forall x, y \in \mathbb{R}^m \quad |F(x) - F(y)| \leq \sigma \|x - y\|_{\ell_2^n}$$

we get

$$\gamma_m\left(\left\{x \in \mathbb{R}^m \mid \left|F(x) - \int F d\gamma_m\right| > t\right\}\right) \leq 2 \exp -K t^2 \sigma^{-2}$$

for a numerical constant K .

The same argument can easily be adapted to other rotational invariant measures than γ_m . For instance, let σ_m be the normalized surface measure on the Euclidean unit sphere $\mathbb{S}^{m-1}$. Using the latter instead of (18) it is possible to prove the following claim. Given $u : \ell_2^n \rightarrow E$, there is an orthonormal basis $x_1, \dots, x_m$ of ℓ_2^n such that the first n vectors $x_1, \dots, x_n$ satisfy

$$\forall \alpha \in \mathbb{R}^n \quad (1 + \epsilon)^{-1/2} \left(\sum_{i=1}^n |\alpha_i|^2 \right)^{1/2} \leq \frac{1}{M} \left\| \sum_{i=1}^n \alpha_i x_i \right\| \leq (1 + \epsilon)^{1/2} \left(\sum_{i=1}^n |\alpha_i|^2 \right)^{1/2},$$

where

$$M = \int \|u(x)\| d\sigma_m(x),$$

and where

$$n = \lfloor \eta'_1(\epsilon) M^2 \|u\|^{-2} \rfloor.$$

This should be compared with the strategy for Pisier's proof of Dvoretzky's Theorem and equation (17). Indeed in the same way, we can choose the orthonormal basis $x_1, \dots, x_m$ at random and show using the above estimate that with positive probability we have the preceding claim.

Corollary 2.12. Let X be as in Theorem 2.6. We have for all $p > 1$

$$\mathbb{E}\|X\| \leq (\mathbb{E}\|X\|^p)^{1/p} \leq K(p)\|X\|,$$

where $K(p)$ is a constant depending only on p .

Proof. Note that, as observed in Remark 2.7,

$$\mathbb{E}|\zeta(X)| = \left(\frac{2}{\pi}\right)^{1/2} (\mathbb{E}|\zeta(X)|^2)^{1/2};$$

hence $\sigma(X) \leq (\pi/2)^{1/2} \mathbb{E}\|X\|$. Assume for simplicity that $\mathbb{E}\|X\| = 1$. Then by Theorem 2.10 we get

$$\mathbb{P}(\{\|X\| > t + 1\}) \leq \exp -Kt^2 \left(\frac{2}{\pi}\right). \quad (24)$$

But

$$\begin{aligned} \mathbb{E}\|X\|^p &= \iint_0^\infty \chi_{\{t < \|X\|^p\}} dt d\omega \\ &= \iint_0^\infty pt^{p-1} \chi_{\{t^p < \|X\|^p\}} dt d\omega \\ &= \int_0^\infty pt^{p-1} \mathbb{P}(\{\|X\| > t\}) dt, \end{aligned}$$

hence (24) implies

$$\mathbb{E}\|X\|^p \leq K(p)^p$$

for some constant $K(p)$ depending only on p . □

Note: By the corollary we get the equivalence of L^p norms of Gaussian vectors, and this will come in handy. We now prove two technical lemmas that will allow us to discretize the problem, permitting us to look at a finite number of directions and X -s, approximating every other one by these ones.

Lemma 2.13. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$ with unit ball B and unit sphere S . Let $\delta > 0$. There is a subset $A \subset S$ which is a δ -net of S with respect to the norm $\|\cdot\|$ (i.e. for any $x \in S$, $\exists y \in A$ s.t. $\|x - y\| < \delta$) with cardinality

$$|A| \leq \left(1 + \frac{2}{\delta}\right)^n.$$

Proof. Let $(y_i)_{i \leq m}$ be a maximal subset of S such that $\|y_i - y_j\| \geq \delta$ for all $i \neq j$. By maximality, $(y_i)_{i \leq m}$ is a δ -net of S . To majorize, we note that the balls $y_i + (\delta/2)B$ are disjoint and included into $(1 + \delta/2)B$. Therefore,

$$\sum_{i \leq m} \text{vol}\left(y_i + \frac{\delta}{2}B\right) \leq \text{vol}\left(\left(1 + \frac{\delta}{2}\right)B\right) = \left(1 + \frac{\delta}{2}\right)^n \text{vol}(B);$$

hence

$$m \left(\frac{\delta}{2}\right)^n \text{vol}(B) \leq \left(1 + \frac{\delta}{2}\right)^n \text{vol}(B),$$

which gives

$$m \leq \left(\frac{2}{\delta} + 1\right)^n$$

as announced. □

Lemma 2.14. For each $\epsilon > 0$ there is a $\delta = \delta(\epsilon)$, $0 < \delta < 1$, with the following property. Let n be any integer and let $\|\cdot\|$ be any norm on $\mathbb{R}^n$. Let A be a δ -net in the unit sphere $(\mathbb{R}^n, \|\cdot\|)$ and let $(B, \|\cdot\|_B)$ be a Banach space, $x_1, \dots, x_n \in B$. Then if

$$\forall \alpha \in A \quad 1 - \delta \leq \left\| \sum_{i=1}^n \alpha_i x_i \right\|_B \leq 1 + \delta, \quad (25)$$

then

$$\forall \alpha \in \mathbb{R}^n \quad (1 + \epsilon)^{-1/2} \|\alpha\| \leq \left\| \sum_{i=1}^n \alpha_i x_i \right\|_B \leq (1 + \epsilon)^{1/2} \|\alpha\|.$$

In particular, let F be the subspace of B generated by $x_1, \dots, x_n$. Then F is $(1 + \epsilon)$ -isomorphic to $(\mathbb{R}^n, \|\cdot\|)$.

Note: We are basically saying that, choosing a finite quantity of vectors, if we are close enough to a sphere on a dense enough δ -net (so on a finite quantity of linear combination of our vectors!) we are close enough to a sphere in every direction in $\mathbb{R}^n$. This means we only need to check a finite quantity of directions to ensure a bound on the distance between two spaces.

The strategy of proof leverages precisely this mechanism: we approximate.

Proof. Assume $\|\alpha\| = 1$. There is $y^0 \in A$ such that $\|\alpha - y^0\| \leq \delta$, hence $\alpha = y^0 + \lambda_1 \alpha'$ with $|\lambda_1| \leq \delta$ and $\|\alpha'\| = 1$. Continuing in the same way, we obtain $\alpha = y^0 + \lambda_1 y^1 + \lambda_2 y^2 + \dots$ with $y_j \in A$ and $|\lambda_j| \leq \delta^j$. It follows that

$$\begin{aligned} \left\| \sum_{i=1}^n \alpha_i x_i \right\|_B &= \left\| \sum_{i=1}^n \sum_{j \geq 0} y_j^i \lambda_j x_i \right\|_B \\ &= \left\| \sum_{j \geq 0} \lambda_j \sum_{i=1}^n y_j^i x_i \right\|_B \\ &\leq \delta^j \left\| \sum_{i=1}^n y_j^i x_i \right\|_B \\ &\leq \sum_{j \geq 0} \delta^j (1 + \delta) \\ &= (1 - \delta)^{-1} (1 + \delta). \end{aligned}$$

Similarly,

$$\begin{aligned} \left\| \sum_{i=1}^n \alpha_i x_i \right\| &\geq \left\| \sum_{i=1}^n y_i^0 x_i \right\| - \left\| \sum_{i=1}^n \sum_{j \geq 1} y_j^i x_i \lambda_j \right\| \\ &\geq 1 - \delta - \delta(1 + \delta)(1 - \delta)^{-1} \\ &= (1 - 3\delta)(1 - \delta)^{-1}, \end{aligned}$$

as $y^0 \in A$ implies we can use the hypothesis (25). If we choose $\delta = \delta(\epsilon)$ small enough so that

$$(3\delta)(1 - \delta)^{-1} \geq (1 + \epsilon)^{-1/2} \text{ and } (1 + \delta)(1 - \delta)^{-1} \leq (1 + \epsilon)^{1/2}$$

then we obtain the result. Note that δ is independent of N ! □

Note: We now have enough tools to pass on and prove Theorem 2.6. We will follow the strategy of proof we discussed.

Proof of theorem 2.6. Let $X_1, \dots, X_n$ be i.i.d. copies of X . We apply the two preceding lemmas with $\|\alpha\| = (\sum_{i=1}^n |\alpha_i|^2)^{1/2}$ (the Euclidean norm on $\mathbb{R}^n$). Assume $\sum_{i=1}^n |\alpha_i|^2 = 1$. Then by Lemma 1.22 the r.v. $\sum_{i=1}^n \alpha_i X_i$ has the same distribution as X . Therefore, we have

$$\mathbb{E} \left\| \sum_{i=1}^n \alpha_i X_i \right\| = \mathbb{E} \|X\|,$$

and by (18) using $t = \delta \mathbb{E} \|X\|$ we get

$$\mathbb{P} \left(\left\| \sum_{i=1}^n \alpha_i X_i \right\| - \mathbb{E} \|X\| > \delta \mathbb{E} \|X\| \right) \leq 2 \exp(-K \delta^2 d(X)) \quad \forall \delta > 0.$$

Let $M = \mathbb{E} \|X\|$ and let A be as in Lemmas 2.13 and 2.14. The preceding inequality implies

$$\begin{aligned} \mathbb{P} \left(\left\{ \exists \alpha \in A \left\| \sum_{i=1}^n \alpha_i X_i \right\| M^{-1} - 1 \right\} > \delta \right) &\leq 2|A| \exp(-K \delta^2 d(X)). \\ &\leq 2 \exp \left\{ \frac{2n}{\delta} - K \delta^2 d(X) \right\}. \end{aligned} \quad (26)$$

Now, if we choose $\delta = \delta(\epsilon)$, the function of ϵ given by the Lemma 2.14 yields the desired result. Indeed, observe that if

$$\frac{2n}{\delta} \leq \frac{1}{2} K \delta^2 d(X), \quad (27)$$

the probability in (26) is not greater than $2 \exp(-(1/2)K \delta^2 d(X))$. Moreover, we can always assume that $d(X)$ is large enough (say $d(X) \geq 4(K \delta^2)^{-1}$), otherwise there is nothing to prove. Hence we can assume the right side of (26) is less than 1. We obtain with positive probability that

$$\left| M^{-1} \left\| \sum_{i=1}^n \alpha_i X_i \right\| - 1 \right| \leq \delta.$$

By Lemma 2.14 (recall we chose $\delta = \delta(\epsilon)$) we conclude that for all α in the sphere of ℓ_2^n we have

$$(1 + \epsilon)^{-1/2} \leq M^{-1} \left\| \sum \alpha_i X_i \right\| \leq (1 + \epsilon)^{1/2},$$

and by homogeneity this implies (with the notation presented in the strategy of proof) that $\mathbb{P}(\Omega(n, \epsilon)) > 0$.

Hence if F_ω is the span of $X_1(\omega), \dots, X_n(\omega)$ and recalling (27), if $n = \lfloor 1/4K\delta(\epsilon)^3 d(X) \rfloor$, we have proved that with positive probability we have $d(F_\omega, \ell_2^n) \leq 1 + \epsilon$. $\square$

Note: We now turn to the proof of Theorem 2.3. For this we will use the following lemma to find a set of vectors that resemble an orthonormal set of vectors in our Banach space. To do this we will have to pay a dimensional price (we cannot expect to find a full basis in general).

Lemma 2.15. Let E be an N -dimensional space. Let $\bar{N} = \lfloor N/2 \rfloor$. Then there are $\bar{N}$ elements $(x_k)_{k \leq \bar{N}}$ in E satisfying

$$\left\| \sum_{i=1}^{\bar{N}} \alpha_i x_i \right\| \leq \left(\sum_{i=1}^{\bar{N}} |\alpha_i|^2 \right)^{1/2} \quad \forall \alpha \in \mathbb{R}^{\bar{N}}, \quad (28)$$

and

$$\|x_k\| \geq \frac{1}{2} \quad \forall k \leq \bar{N}. \quad (29)$$

Proof. Let $\alpha(u) = \|u\|$. By Theorem 1.20, there is an isomorphism $u : \ell_2^n \rightarrow E$ such that $\alpha(u) = 1$ and $\alpha^*(u^{-1}) = N$. We claim that this implies

$$\alpha_k(u) \geq 1 - \frac{k}{N} \text{ for } k = 1, 2, \dots, N.$$

Indeed, let P be any orthogonal projection on ℓ_2^n with $\text{rank } P < k$. Then

$$\begin{aligned} N - k &< \text{rank}(I - P) \\ &= \text{tr}(I - P) \\ &= \text{tr } u^{-1}u(I - P) \\ &\leq \alpha^*(u^{-1})\|u(I - P)\| \\ &\leq N\|u - uP\|. \end{aligned}$$

Where each of these passages is justified by $I - P$ being also a projection onto the complementary space, basic properties of projections and. We therefore get $\|u - uP\| > 1 - k/N$, hence $\alpha_k(u) \geq 1 - k/N$. By Lemma 1.18, there is an orthonormal basis $f_1, \dots, f_N$ of ℓ_2^n such that

$$\|u(f_k)\| \geq \alpha_k(u) \geq 1 - k/N \quad (30)$$

for all k . Let $x_k = u(f_k)$. Then (since (f_k) is orthonormal and $\|u\| \leq 1$) we get

$$\begin{aligned} \left\| \sum_{i=1}^N \alpha_i x_i \right\| &= \left\| \sum_{i=1}^N u(f_i) \alpha_i \right\| \\ &\leq \|u\| \left| \sum_{i=1}^N f_i x_i \right| \\ &\leq \left| \sum_{i=1}^N f_i x_i \right| \\ &= \left(\sum_{i=1}^n |\alpha_i|^2 \right)^{1/2}. \end{aligned}$$

So a fortiori (28) is satisfied. On the other hand, if $k \leq \lfloor N/2 \rfloor$, by (30) we get $\|x_i\| \geq 1/2 \ \forall i = 1, \dots, N/2$. $\square$

Note: We used the map u , which is continuous with continuous inverse, to bring out the geometry of ℓ_2^n onto E . This is not "for free", we have to pay a dimensional price if we want to ensure that the vectors don't end up being zero, but even though here we do this with $\lfloor N/2 \rfloor$ it can be done with any integer strictly smaller than N . As close as we get to N , we will get a weaker bound on the norms of the x_i -s, but any uniform bound is enough to prove Theorem 2.3.

We need one last Lemma before proving Theorem 2.3, which is a technical lemma that lets us estimate the expected value of the sup of independent Gaussian variables.

Lemma 2.16. Let $1 \leq q < \infty$. Let $\gamma(q) = (\mathbb{E}|g|^q)^{1/q}$ for g Gaussian random variable. Let $Z_1, \dots, Z_N$ be N real-valued Gaussian variables (with mean zero but arbitrary variance). Then

$$\mathbb{E} \left(\sum_{i=1}^N |Z_k|^q \right)^{1/q} \leq \left(\sum_{i=1}^N \mathbb{E}|Z_k|^q \right)^{1/q} \leq \gamma(q) N^{1/q} \sup_{k \leq N} \|Z_k\|_2, \quad (31)$$

and

$$\mathbb{E} \sup_{k \leq N} |Z_k| \leq c(1 + \log N)^{1/2} \sup_{k \leq N} \|Z_k\|_2, \quad (32)$$

where c is a numerical constant. On the other hand, for independent standard variables $g_1, \dots, g_N$, we have

$$\left(\frac{2}{\pi}\right)^{1/2} N^{1/q} \leq \mathbb{E} \left(\sum_{i=1}^N |g_i|^q \right)^{1/q} \quad (33)$$

and

$$c'(\log N)^{1/2} \leq \mathbb{E} \sup_{k \leq N} g_k \leq \sup_{k \leq N} |g_k| \leq \mathbb{E} \sup_{k \leq N} |g_k|. \quad (34)$$

Proof. A Gaussian variable Z_k is a multiple of a standard Gaussian variable. Hence, the distribution is the same as that of $\|Z_k\|_2 g_1$ so that $\|Z_k\|_q = \|Z_k\|_2 \gamma(q)$ for each k . Let

$$Y = \left(\sum_{i=1}^N |Z_k|^q \right)^{1/q}.$$

Using $\|Y\|_1 \leq \|Y\|_q$, (31) follows, as

$$\begin{aligned} \mathbb{E} \left(\sum_{i=1}^N |Z_k|^q \right)^{1/q} &= \|Y\|_1 \leq \|Y\|_q = \left(\mathbb{E} \sum_{i=1}^N |Z_k|^q \right)^{1/q} \\ &= \left(\sum_{i=1}^N \|Z_k\|_q^q \right)^{1/q} \\ &= \left(\|Z_k\|_2^q \gamma(q)^q \right)^{1/q} \\ &\leq \gamma(q) N^{1/q} \sup_{k \leq N} \|Z_k\|_2. \end{aligned}$$

For some constant K we have that $\gamma(q) \leq K q^{1/2} \forall q \geq 1$ ((13)). Therefore, by (31),

$$\begin{aligned} \mathbb{E} \sup_{k \leq N} |Z_k| &\leq \left(\mathbb{E} \sum_{k \leq N} |Z_k|^q \right)^{1/q} \\ &\leq \left(\mathbb{E} \sum_{k \leq N} |Z_k|^q \right)^{1/q} \\ &\leq K q^{1/2} N^{1/q} \sup_{k \leq N} \|Z_k\|_2. \end{aligned}$$

Hence, by taking $q = 1 + \log N$, (32) follows, as

$$N^{1/q} = e^{\frac{\log N}{1 + \log N}} < e.$$

The inequality (33) comes from

$$\begin{aligned} \left(\frac{2}{\pi}\right)^{1/2} N^{1/q} &= \gamma(1) N^{1/q} \\ &= \mathbb{E} |g_1| N^{1/q} \\ &\leq (\mathbb{E} |g_1|^q)^{1/q} N^{1/q} \\ &= \mathbb{E} \left(\sum_{i=1}^N |g_i|^q \right)^{1/q}. \end{aligned}$$

By Lemma 1.23,

$$\mathbb{P}\left(\left\{\sup_{k \leq N} g_k \leq \alpha(\log N)^{1/2}\right\}\right) \leq (1 - 1/N)^N \leq e^{-1},$$

from which we can derive (34), as from

$$\sup_{k \leq N} g_k \geq \chi_{\{\sup_{k \leq N} g_k \geq \alpha(\log N)^{1/2}\}} \alpha(\log N)^{1/2}$$

we get

$$\mathbb{E} \sup_{k \leq N} g_k \geq (1 - e^{-1}) \alpha(\log N)^{1/2},$$

as desired. $\square$

Proof of theorem 2.3: Let $(g_k)_{k \in \mathbb{N}}$ be a sequence of i.i.d. standard Gaussian r.v.'s as before. Let then $(x_k)_{k \leq \bar{N}}$ be as in Lemma 2.15. Let

$$X = \sum_{k \leq \bar{N}} g_k x_k.$$

We claim

$$d(X) \geq c \log N \tag{35}$$

for some numerical constant $c > 0$. Using (35) and applying Theorem 2.6, we immediately obtain Theorem 2.3.

It remains to prove (35): first note that (28) implies $\sigma(X) \leq 1$, as (see Theorem 2.10 and Lemma 2.15) $\sigma(X) = \|u\| \leq 1$. On the other hand, we have that

$$\mathbb{E} \sup_{k \leq \bar{N}} |g_k| \|x_k\| \leq \mathbb{E} \|X\|. \tag{36}$$

Now, (29) gives

$$\frac{1}{2} \mathbb{E} \sup_{k \leq \bar{N}} |g_k| \leq \mathbb{E} \|X\|,$$

so by Lemma 2.16, we get

$$c'(\log \bar{N})^{1/2} \leq \mathbb{E} \sup_{k \leq \bar{N}} |g_k|$$

therefore we conclude (35) holds.

Note: We wanted to exhibit an X with an high density of dimension as discussed in the strategy of proof. As one could expect, we chose the vectors we had built in Lemma 2.16 to find a linear combination of vectors that were as perpendicular as possible, hence maximizing $d(X)$.

Note that the estimate is sharp: for $E = \ell_\infty^N$, we have that $\delta(E) \sim \log N$, and moreover whenever $n > c \log N$ for some numerical constant c , we don't have any sections. However, whenever we are far from ℓ_∞^N , we can find better estimates (see [4], [5]).

Theorem 2.17. There are functions $\alpha(\epsilon) > 0$ and $\beta(\epsilon) > 0$ such that for all $N > 1$ we have

$$\alpha(\epsilon) \log N \leq n_\epsilon(\ell_\infty^N) \leq \beta(\epsilon) \log N.$$

Moreover for each $2 \leq q < \infty$, there are functions $\alpha(q, \epsilon) > 0$ and $\beta(q, \epsilon) > 0$ such that for all $N > 1$,

$$\alpha(q, \epsilon) N^{2/q} \leq n_\epsilon(\ell_q^N) \leq \beta(q, \epsilon) N^{2/q},$$

whereas for each $1 \leq q \leq 2$, there is a function $\phi(\epsilon)$ such that for all $N > 1$,

$$\phi(\epsilon) N \leq n_\epsilon(\ell_q^N) \leq N.$$

Proof. Fix q such that $1 \leq q \leq \infty$. Let (e_k) be the canonical basis of ℓ_q^N and let $X = \sum_{i=1}^N g_i e_i$. If we consider X as a random variable with values into ℓ_q^N . by Lemma 2.16 we get

$$d(X) \sim \begin{cases} \log N, & \text{for } q = \infty; \\ N^{2/q}, & \text{for } 2 \leq q < \infty; \\ N, & \text{for } 1 \leq q \leq 2. \end{cases} \quad (37)$$

This comes from $\sigma(X) \leq 1$ for $2 \leq q \leq \infty$ and $\sigma(X) \leq N^{1/q-1/2}$ for $1 \leq q \leq 2$. $\square$

Bibliography

- [1] Aryeh Dvoretzky. Some results on convex bodies and banach spaces. In *Proceedings of the International Symposium on Linear Spaces (Jerusalem, 1960)*, pages 123–160, Jerusalem / Oxford, 1961. Academic Press / Pergamon Press.
- [2] T. Figiel. A short proof of dvoretzky's theorem. *Compositio Mathematica*, 33(3):297–301, 1976.
- [3] V. D. Milman. New proof of the theorem of A. Dvoretzky on sections of convex bodies. *Funkcional. Anal. i Priložen.*, 5(4):28–37, 1971.
- [4] Vitali D. Milman and Gideon Schechtman. *Asymptotic Theory of Finite Dimensional Normed Spaces*, volume 1200 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, Heidelberg, 1986. With an Appendix by M. Gromov: "Isoperimetric Inequalities in Riemannian Manifolds".
- [5] Gilles Pisier. *The Volume of Convex Bodies and Banach Space Geometry*, volume 94 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 1989.
- [6] A. Szankowski. On dvoretzky's theorem on almost spherical sections of convex bodies. *Israel Journal of Mathematics*, 17:325–338, 1974.